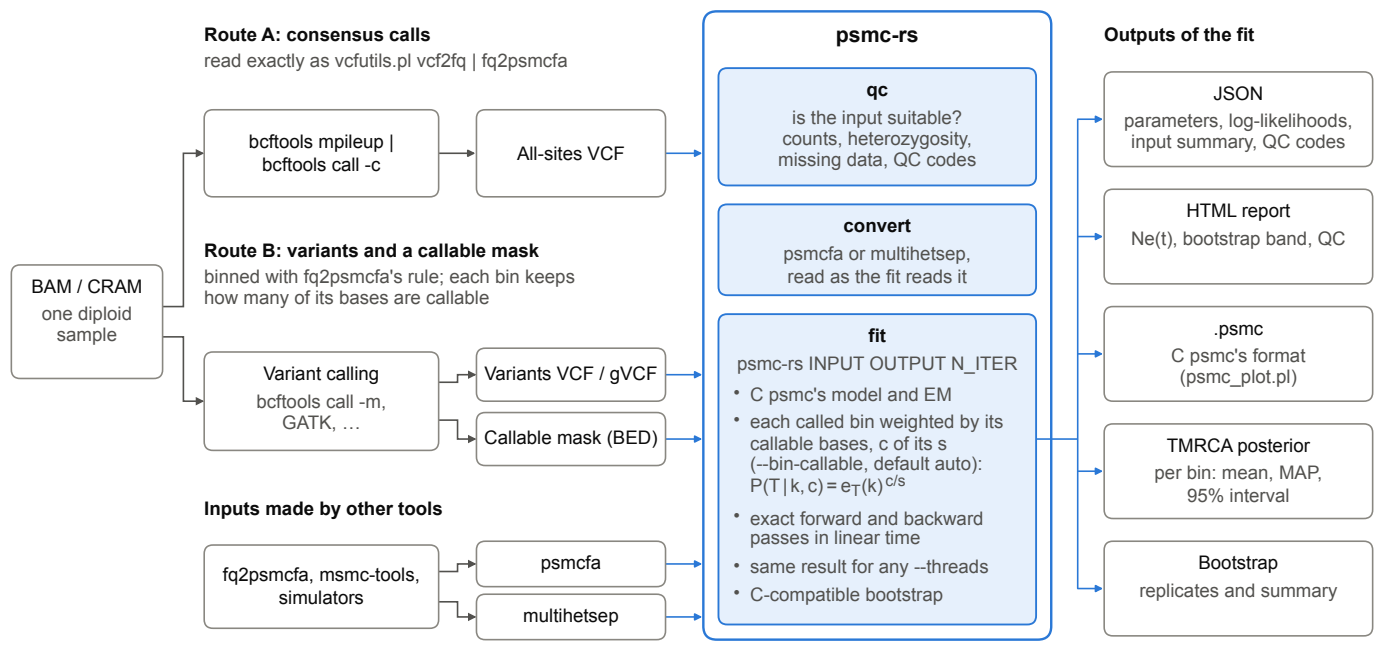
Route A: consensus calls
read exactly as vcfutils.pl vcf2fq | fq2psmcfa
bcftools mpileup | bcftools call -c
All-sites VCF
BAM / CRAM
one diploid sample
Route B: variants and a callable mask
binned with fq2psmcfa's rule; each bin keeps how many of its bases are callable
Variant calling
bcftools call -m, GATK, …
Variants VCF / gVCF
Callable mask (BED)
Inputs made by other tools
fq2psmcfa, msmc-tools, simulators
psmcfa
multihetsep
psmc-rs
qc
is the input suitable? counts, heterozygosity, missing data, QC codes
convert
psmcfa or multihetsep, read as the fit reads it
fit
psmc-rs INPUT OUTPUT N_ITER
C psmc's model and EM
each called bin weighted by its callable bases, c of its s (--bin-callable, default auto): P(T | k, c) = e_T(k)^c/s
exact forward and backward passes in linear time
same result for any --threads
C-compatible bootstrap
Outputs of the fit
JSON
parameters, log-likelihoods, input summary, QC codes
HTML report
Ne(t), bootstrap band, QC
.psmc
C psmc's format (psmc_plot.pl)
TMRCA posterior
per bin: mean, MAP, 95% interval
Bootstrap
replicates and summary